

RB005 Week 3 — English Worksheet

Topic: Core Mechanic — Pick Your Pattern · **Course:** Dream Game Studio · **Time:** about 45 minutes

You finished RB001–004, so you own a toolbox of script **patterns**. This week you pick the right pattern from the toolbox and script your game's **core mechanic** — the thing the player does again and again.

1 · Predict

Read each pattern. Before you run it in Studio, write what you think will happen.

Pattern A — the pickup

```
local coin = script.Parent
local debounce = false

coin.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player and not debounce then
        debounce = true
        player.leaderstats.Coins.Value += 1
        coin:Destroy()
    end
end)
```

A player touches the coin. What happens to their leaderstats? Why does the coin only count once?

Pattern B — the countdown

```

local timeLeft = 30

while timeLeft > 0 do
    print(timeLeft .. " seconds left!")
    task.wait(1)
    timeLeft -= 1
end

print("Time is up!")

```

What prints in the Output, and how long does the whole loop take?

Pattern C — the if-gate

```

local door = workspace.GoldDoor

door.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player and player.leaderstats.Coins.Value >= 10 then
        door.CanCollide = false
        door.Transparency = 0.5
    end
end)

```

A player with 4 coins touches the door. What happens? What about a player with 12 coins?

Match the pattern to the game idea. Write A, B, or C next to each, then say which pattern is closest to YOUR game's core mechanic and why.

A collecting game where players grab gems for points → ____

A survival game where each round lasts 30 seconds → ____

An adventure game with a boss door that needs 10 keys → ____

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — One touch should give the player **1** coin, then the coin disappears. Right now one touch gives 5, 10, sometimes 20 coins.

```
local coin = script.Parent

coin.Touched:Connect(function(hit)
    local player = game.Players:GetPlayerFromCharacter(hit.Parent)
    if player then
        player.leaderstats.Coins.Value += 1
    end
end)
```

Hint: `Touched` fires many times for one touch. What did we use in RB003 to stop that?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Lava should set the toucher's health to 0. It works for players, but the Output shows a red error when a loose part falls onto the lava.

```
local lava = script.Parent

lava.Touched:Connect(function(hit)
    local humanoid = hit.Parent:FindFirstChild("Humanoid")
    humanoid.Health = 0
end)
```

Hint: a falling part has no Humanoid — so what is `humanoid` then?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The countdown should tick once per second. Right now all 30 numbers print instantly in one frame.

```
local timeLeft = 30

while timeLeft > 0 do
    print(timeLeft .. " seconds left!")
    timeLeft -= 1
end
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your game and script YOUR core mechanic, using the pattern you matched in section 1 (or a mix). When it works, come back here.

Send a photo or video of your mechanic working, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My core mechanic is ...

I used the ... pattern because ...

The event my script listens for is ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while your core mechanic runs. Talk through it like you are teaching someone who has never seen your game. Try to use these words: **pattern**, **event**, **debounce**, **leaderstats**, **task.wait**.

1. Play your game and show the core mechanic working.
2. Name the pattern you used from the toolbox.
3. Open your script and read the most important line out loud.
4. Explain one check in your code (like `if player` or `not debounce`) and why it is there.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.